Tire Pressure Monitoring System.

1. In the siemens VDO Tire Guard System, actual air pressure in the tyre, is measured with a micro-mechanical sensor, which is silicon based vaccum measuring cell. The sensor is diaphragm-based and is embedded in a vibration absorbing compound in the plastic housing which is screwed into the wheel bed with the valve nut. The diaphragm-deformation under different air pressure is measured using piezoesensitive sensors and processed electronically.
The measurement values obtained by the sensor are transmitted at a frequency of about 315 MHz in USA and 433 MHz in Europe by radio, to the controller. Any drop in pressure in indicated in the form of a red bulb flashing or an acoustic alarm.

2. As the method is highly accurate, it is possible to detect very small changes in air pressure inespective of vehicle load, degree of tyre wear, temp. of tyre or road condition. Besides, the signal is available even when the vehicle is parked.

3. Deflection Detection System [DDS] of the Continental Automotive systems, however, does not require additional sensors or hardware. Data collected from EBS wheel speed sensors are used for sensing changes in the dynamic rolling radius to detect loss of tyre pressure

4. Continental DDS Plus Software module also detects the vibration characteristics of the tyr if its pressure drops, thereby locating the aff tyre. These Systems can monitor all the four tyres simultaneously, as in direct measuring sys

5. The continental direct measuring system consists 4 intelligent tyre modules and a central cont unit with receiver.

6. The intelligent tyre module contains a pressure and temperature sensor, transmitter and data memory for many functions such as measureme of tyre pressure and temperature, automatic loac detection, storage of tyre data, autolocation funct etc The module is attached to the inside of the ty as shown. Compared to the usual placement of sensors on the wheel rim, this arrangemen has the following advantages:-

(1) The module is far less likely to be damagec during fitting of the tyre.

(2) Measurement of tyre temperature is more accu since problematic incorrect measurements at the wheel rim due to sun, snow or hot brake disc are ruledout.

Self Inflating Tyres.

1. Introduction:-

There are three general goals of a tyre-inflation system:

(1) To detect when the air pressure in a particular tyre has dropped, by continous monitoring of air pressure in each tyre.

(2) Notify the driver of the problem.

(3) Inflate the affected tyre back to the proper level.

2. Components:-

Main Elements of a tyre inflation system are:-

(1) A valve to isolate individual tyres so that air loss not flow from or to other tyres when one tyre is being checked or inflated.

(2) An air source, usually an existing onboard source example:- braking or pneumatic systems. However, in such a case, there have to be safety checks also to ensure that there is enough air pressure for the primary function of the source before taking air ~~from~~ for tyre-inflation.

(3) A means to ~~t~~ send the air from the source to the tyres. This is usually done through the axle.

(4) A pressure-relief vent to vent the air from the tyre without any damage to the hub or rear-axle seals.

(3) Currently available systems:-

The main system currently in use is the Centr Tyre Inflation System [CTIS] which is bein manufactured by two main manufactures:-

(1) US based Dana Corporation, having two versions:- (i) CTIS for millitary use
(ii) The tyre pressure Control System (TPCS for commercial, heavy machinery use.

(2) France-based Syegon, which is division of GIAT.

Brief Working of Central Tyre Inflation System:-

1. A wheel valve is located at each wheel end. In case of dual wheels, valves are connected only to the outer wheels for balancing the pressure between the two wheels. The wheel enables on-demand inflation and deflection of the tyres. It also isolates the tyre from the system when it is not in use so as to release the pressure on the seal and extend its life.
2. The electronic Control Unit (ECU) mounted behind the ~~passage~~ passenger seat processes the driver commands, monitors all signals throughout the system and commands the system to check tyre pressure every 10 minutes to make sure that the desired pressure is being maintained.
3. ECU sends a command to pneumatic control unit, which directly controls the wheel valves and air system. The pneumatic control unit also has a sensor to transmit the tyre pressure readings to the ECU.
4. A dash-mounted operator control panel shows a current tyre pressures, selected modes and system status. It also permits the driver to select the tyre-pressure modes to match current conditions. On selecting a tyre-pressure setting by the driver, signal from the control panel is sent to the ECU, which passes the same to the pneumatic control unit and to the wheel valves.

5. At higher speeds, such as a highways, tyre pressure ~~measured~~ should be higher to pr tyre damage. The CTIS has a speed sensor u sends speed information to the ECU and if th vehicle continues to move at higher speed fo given set period of time, the system automat inflates tyres to the desired pressure for tha speed.

6. This system uses air from the same compressor which supplies air to the braking system. A pressure switch ensures that the braking system gets priority by preventing the CTIS from takin air from the supply tank till the braking system is fully charged.

Under Development Systems.

The efforts of the two companies to develop a real self-inflating tyre, without taking compressed air from an on-board external source, are worth mension.

1. The Czech Republic based company, 'Coda Development' has patented their self Inflating Tyre (SIT) system and have already received a Tire Technology International Award for Innovation and Excellence for its SIT at the 2008 SAE World Congress in 2008.

According to the company, "The SIT system is based on highly reliable and proven peristaltic pump principles. It uses the weight and motion of the vehicle to inflate the tyre as needed, sourcing air from the outside atmosphere. The whole system consits of only two components - a tube chamber functioning as a peristaltic pump for the tyre and a pressure management device to control the inflation.

Figure B :- Internal Air Circulation in the Correctly inflated tyre of the SIT system.

Figure C :- Under Inflated tyre being inflated by the SIT System.

- The peristaltic tubing is located logitudinal between the rim and the tyre wall and follo almost the whole perimeter of the rim.
- Normal tyre deformation caused by the we of the vehicle causes a closure of the tubi at its lowest point.
- As the tyre moves against the road, this closure pushes the air contained inside the tubing into the tyre and simultaneously it p outside ~~the~~ air back into the tubing. As a resu the tyre is inflated with the contents of the tubing with each wheel revolution until it rea its desired pressure."

- Figure B:- Figure B shows the tyre, the rim a tubing between them forming the SIT syste The wheel rotation indicated by an arrow squeezes the SIT tubing at the wheel bo producing vaccum at space A and pressure at space B, which causes the air through the tubi to be forced into the tyre and also the air fro the tyre into the tubing through pressure measurement device (which can be pre-se at the device or alternately, by the driver). In this way, the air keeps circulating betwe the tyre and the tubing as shown by arrow

In case of the under-inflated tyre [figure c], the pressure management device closes the intake opening from the tyre due to which the created vaccum at space A opens the check valve. This draws air from the atmosphere into the tyre as shown by arrow.
When the tyre gets inflated correctly, the pressure management device again opens, destroying the vaccum at A due to which the check valve is closed again and the internal air circulation restarts.

If the pressure management device is electronically manged, it can be connected to the vehicle's TPMS. In such case the TPMS, alongwith SIT, would not only inform the driver about the under-inflated tyres, but would also start the inflation immediately and solve the problem as soon as it ~~happened~~ appeared.
The company aims to bring this technology to the market very soon.
Tests conducted by the company on a regular passenger car wheel have proven that the forces between the deformed tyre wall and the rim are sufficient to generate significantly higher pressure than what is needed for tyre inflation.

2. Goodyear has also displayed their 'Air Maintainance Technology' (AMT) at the Geneva Motor Show 2012. AMT has been under development in Goodyear laboratories for the past two years and is claimed to require no external pump, electronics or drive intervention

and is promoted as a technology that dispe
with any need to check tyre pressure. Howeve
no further details regarding technology or expect
production target has been given by the company